

Assignment - 18

Requirements:

```
create table ENROLLED_STUDENTS(  
  StudentID int primary key,  
  FirstName VARCHAR(50) NOT NULL,  
  LastName VARCHAR(50) NOT NULL,  
  BirthDate DATE,  
  PhoneNumber VARCHAR(15));
```

```
CREATE TABLE COURSES (CourseID INT PRIMARY KEY,  
  CourseName VARCHAR(100) NOT NULL,  
  Credits int NOT NULL CHECK (Credits > 0));
```

```
CREATE TABLE ENROLLMENTS(  
  EnrollmentID INT PRIMARY KEY,  
  StudentID INT,  
  CourseID INT,  
  CONSTRAINT fk_student  
    Foreign KEY (StudentID)  
    REFERENCES ENROLLED_STUDENTS(StudentID)  
    ON DELETE CASCADE,  
  CONSTRAINT fk_course  
    FOREIGN KEY (CourseID)  
    REFERENCES COURSES(CourseID)  
    ON DELETE CASCADE);
```

1. **Insert a Record into ENROLLED_STUDENTS:** Write a query to insert a new student with the following details: StudentID = 101, FirstName = 'John', LastName = 'Doe', BirthDate = '1999-12-01', and PhoneNumber = '123-456-7890'.

-- 1Q

```
INSERT INTO ENROLLED_STUDENTS(StudentID, FirstName, LastName,  
  BirthDate, PhoneNumber) VALUES  
(101, 'John', 'Doe', '1992-12-01', '123-456-7890');
```

2. **Update a Record in ENROLLED_STUDENTS:** Write a query to update the PhoneNumber of the student with StudentID = 101 to '987-654-3210'.

-- 2Q

```
UPDATE ENROLLED_STUDENTS  
SET PhoneNumber = '987-654-3210'  
WHERE StudentID = 101;
```

```
SELECT * FROM ENROLLED_STUDENTS;
```

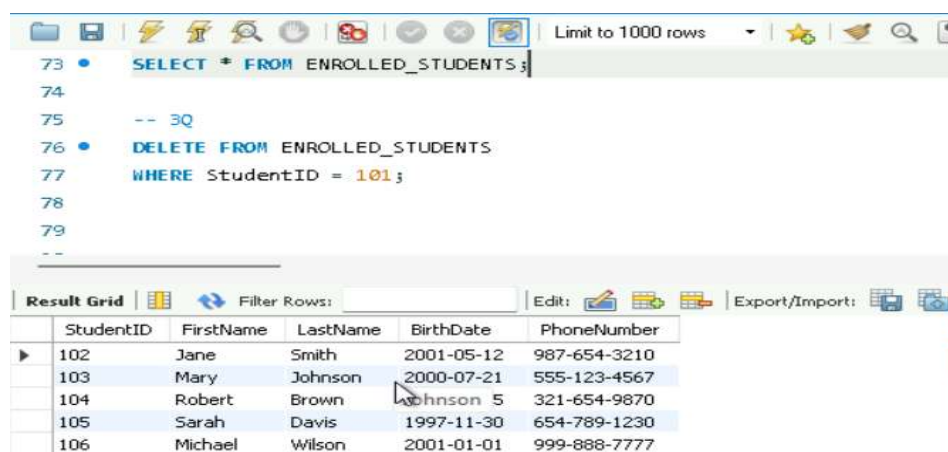


	StudentID	FirstName	LastName	BirthDate	PhoneNumber
▶	101	John	Doe	1992-12-01	987-654-3210
	102	Jane	Smith	2001-05-12	987-654-3210
	103	Mary	Johnson	2000-07-21	555-123-4567
	104	Robert	Brown	1998-03-15	321-654-9870
	105	Sarah	Davis	1997-11-30	654-789-1230

3. Delete a Record from ENROLLED_STUDENTS: Write a query to delete the student with StudentID = 101 from the ENROLLED_STUDENTS table.

-- 3Q

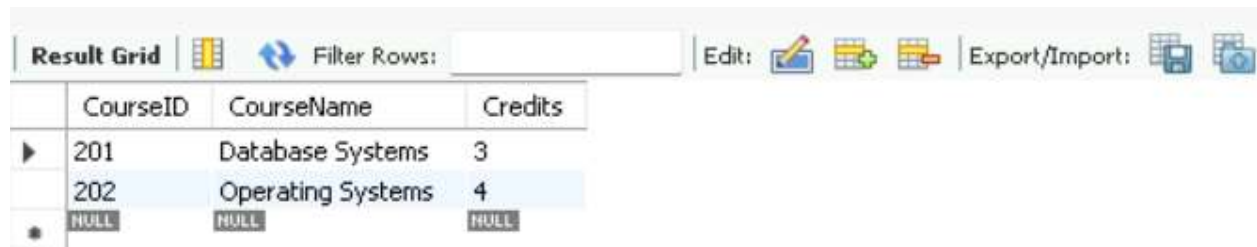
```
DELETE FROM ENROLLED_STUDENTS  
WHERE StudentID = 101;  
SELECT * FROM ENROLLED_STUDENTS;
```



	StudentID	FirstName	LastName	BirthDate	PhoneNumber
▶	102	Jane	Smith	2001-05-12	987-654-3210
	103	Mary	Johnson	2000-07-21	555-123-4567
	104	Robert	Brown	1998-03-15	321-654-9870
	105	Sarah	Davis	1997-11-30	654-789-1230
	106	Michael	Wilson	2001-01-01	999-888-7777

4. Write a query to insert the following courses into the COURSES table: CourseID = 201, CourseName = 'Database Systems', Credits = 3 CourseID = 202, CourseName = 'Operating Systems', Credits = 4

```
INSERT INTO COURSES(CourseID, CourseName, Credits) VALUES
(201, 'Database Systems', 3),
(202, 'Operating Systems', 4));
```

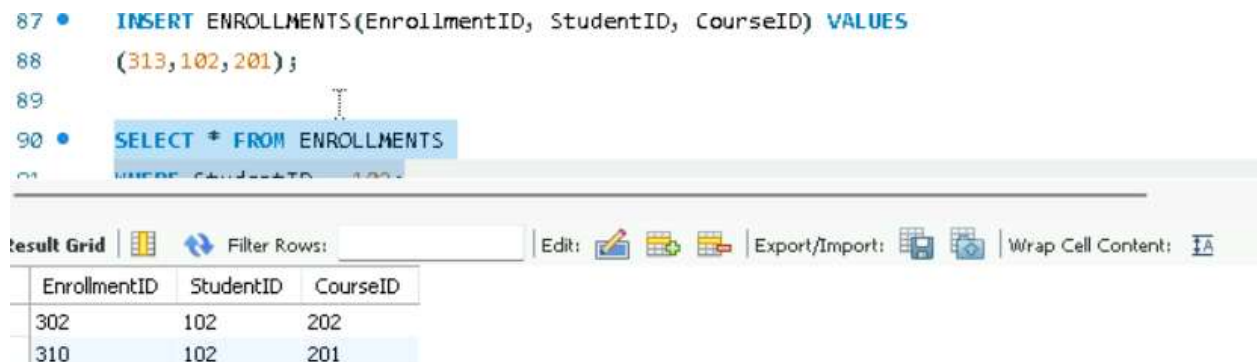


	CourseID	CourseName	Credits
▶	201	Database Systems	3
	202	Operating Systems	4
*	NULL	NULL	NULL

5. Insert Records into ENROLLMENTS with Foreign Keys: Write a query to enroll the student with StudentID = 102 in the course CourseID = 201 and another query to enroll the same student in the course CourseID = 202.

-- 5Q

```
INSERT ENROLLMENTS(EnrollmentID, StudentID, CourseID) VALUES
(313,102,201),
(314,102,202);
```



```
87 • INSERT ENROLLMENTS(EnrollmentID, StudentID, CourseID) VALUES
88   (313,102,201);
89
90 • SELECT * FROM ENROLLMENTS
```

EnrollmentID	StudentID	CourseID
302	102	202
310	102	201

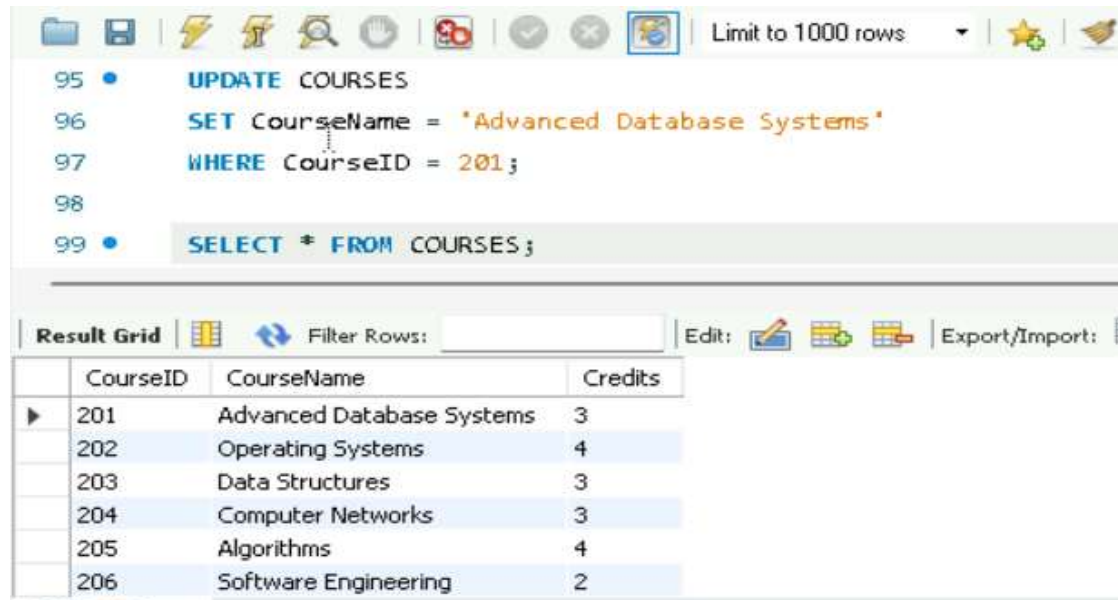
6. Write a query to update the CourseName of CourseID = 201 to 'Advanced Database Systems'.

-- 6Q

```
UPDATE COURSES
```

```
SET CourseName = 'Advanced Database Systems'
WHERE CourseID = 201;
```

```
SELECT * FROM COURSES;
```



The screenshot shows a database query editor with a toolbar at the top. The query text area contains the following SQL statements:

```
95 • UPDATE COURSES
96 SET CourseName = 'Advanced Database Systems'
97 WHERE CourseID = 201;
98
99 • SELECT * FROM COURSES;
```

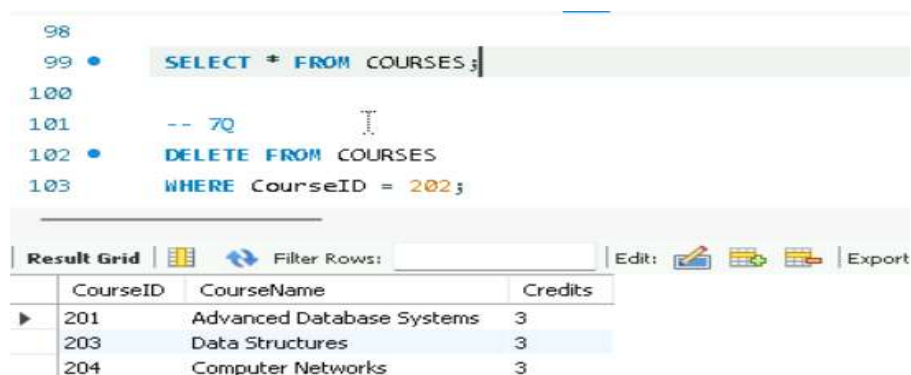
Below the query text is the 'Result Grid' section, which displays the results of the SELECT statement. The grid has columns for CourseID, CourseName, and Credits. The data is as follows:

CourseID	CourseName	Credits
201	Advanced Database Systems	3
202	Operating Systems	4
203	Data Structures	3
204	Computer Networks	3
205	Algorithms	4
206	Software Engineering	2

7. Write a query to delete the course with CourseID = 202 from the COURSES table.

-- 7Q

```
DELETE FROM COURSES
WHERE CourseID = 202;
```



The screenshot shows a database query editor with a toolbar at the top. The query text area contains the following SQL statements:

```
98
99 • SELECT * FROM COURSES;
100
101 -- 7Q
102 • DELETE FROM COURSES
103 WHERE CourseID = 202;
```

Below the query text is the 'Result Grid' section, which displays the results of the SELECT statement. The grid has columns for CourseID, CourseName, and Credits. The data is as follows:

CourseID	CourseName	Credits
201	Advanced Database Systems	3
203	Data Structures	3
204	Computer Networks	3

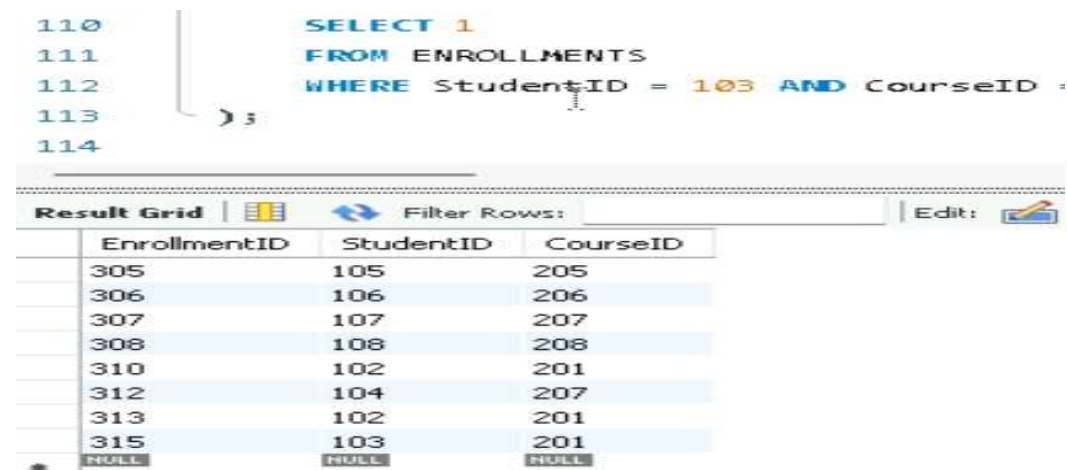
8. Write a query to insert a new enrollment for StudentID = 103 in CourseID = 201, but only if that student isn't already enrolled in the course.

-- 8Q

```
INSERT INTO ENROLLMENTS (EnrollmentID, StudentID, CourseID)
SELECT 315, 103, 201
FROM DUAL
WHERE NOT EXISTS (
    SELECT 1
    FROM ENROLLMENTS
    WHERE StudentID = 103 AND CourseID = 201
);
```

select * FROM ENROLLMENTS;

```
110      SELECT 1
111      FROM ENROLLMENTS
112      WHERE StudentID = 103 AND CourseID =
113      );
114
```



EnrollmentID	StudentID	CourseID
305	105	205
306	106	206
307	107	207
308	108	208
310	102	201
312	104	207
313	102	201
315	103	201
NULL	NULL	NULL

9. Write a query to retrieve the details of all students enrolled in the course with CourseID = 201.

-- 9Q

-- Retrieve complete profile details of students taking Course 201

```
SELECT s.StudentID, s.FirstName, s.LastName, s.BirthDate, s.PhoneNumber
FROM ENROLLED_STUDENTS s
JOIN ENROLLMENTS e ON s.StudentID = e.StudentID
WHERE e.CourseID = 201;
```

```

117      -- 9Q
118      -- Retrieve complete profile details of students taking Course 201
119      • SELECT s.StudentID, s.FirstName, s.LastName, s.BirthDate, s.PhoneNumber
120      FROM ENROLLED_STUDENTS s
121      JOIN ENROLLMENTS e ON s.StudentID = e.StudentID
122      WHERE e.CourseID = 201;
123
124

```

Result Grid					
Filter Rows:		Export:		Wrap Cell Content: ☐	
StudentID	FirstName	LastName	BirthDate	PhoneNumber	
102	Jane	Smith	2001-05-12	987-654-3210	
102	Jane	Smith	2001-05-12	987-654-3210	
103	Mary	Johnson	2000-07-21	555-123-4567	

10. Write a query to retrieve all the courses from the COURSES table that have more than 3 credits.

```

-- 10Q
-- Retrieve all details for courses worth more than 3 credits
SELECT CourseID, CourseName, Credits
FROM COURSES
WHERE Credits > 3;

```

```

124      -- 10Q
125      -- Retrieve all details for courses worth more than 3 credits
126      • SELECT CourseID, CourseName, Credits
127      FROM COURSES
128      WHERE Credits > 3;
129

```

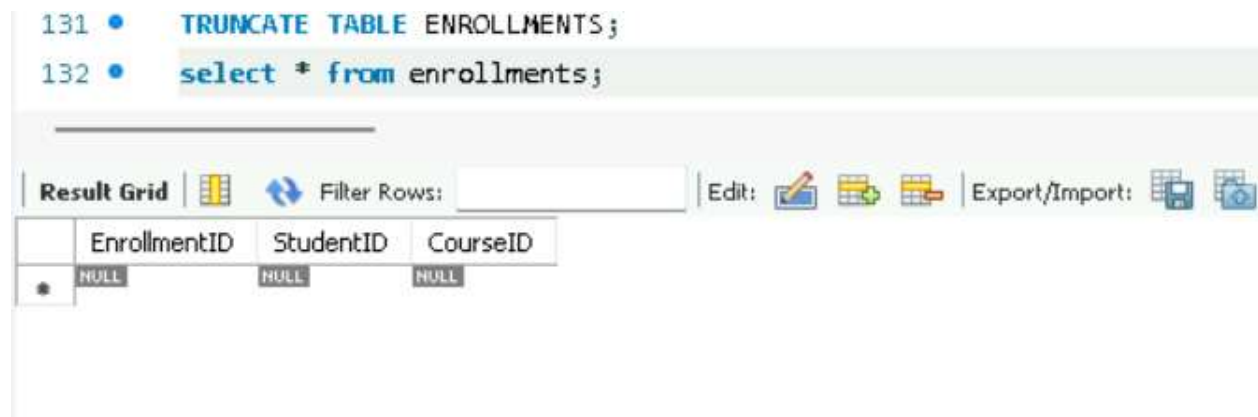
Result Grid			
Filter Rows:		Edit:   	
		Export/Import: 	
CourseID	CourseName	Credits	
205	Algorithms	4	
208	Machine Learning	5	
NULL	NULL	NULL	

11. Write a query to truncate all the data in the ENROLLMENTS table without deleting the table itself.

-- 11Q

```
TRUNCATE TABLE ENROLLMENTS;
```

```
select * from enrollments;
```



The screenshot shows a SQL query editor with two lines of code:

```
131 • TRUNCATE TABLE ENROLLMENTS;  
132 • select * from enrollments;
```

Below the editor is a toolbar with icons for "Result Grid", "Filter Rows", "Edit", and "Export/Import". The "Result Grid" icon is active, and a table with the following data is displayed:

	EnrollmentID	StudentID	CourseID
•	NULL	NULL	NULL